

DAKHLA, Egypt

WMO: 624320

Lat: 25.5028N

Lon: 28.9725E

Elev: 365

StdP: 14.50

Time Zone: 2.00 (E02)

Period: 94-18

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 40.2	(c) 42.3	(d) 25.3	(e) 19.6	(f) 55.9	(g) 28.1	(h) 22.4	(i) 54.6	(j) 16.6	(k) 67.3	(l) 15.0	(m) 65.9	(n) 2.0	(o) 270	(p) 0.392

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 27.7	(c) 109.3	(d) 71.7	(e) 107.3	(f) 70.9	(g) 105.4	(h) 70.4	(i) 74.0	(j) 103.6	(k) 72.8	(l) 102.4	(m) 71.8	(n) 101.3	(o) 7.2	(p) 340

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
63.4	88.7	90.4	61.5	82.7	90.0	60.0	78.4	89.7	37.7	104.5	36.6	102.7	35.7	101.1	82.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
16.3	14.1	12.2	DB	35.5	112.7	2.3	1.9	33.9	114.1	32.5	115.2	31.2	116.2	29.5	117.6	
			WB	32.0	76.9	2.3	2.0	30.3	78.3	29.0	79.4	27.7	80.5	26.0	81.9	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	75.9	58.0	60.9	68.1	76.1	83.6	88.5	90.2	90.4	87.0	79.2	68.3	60.0
	DBStd	12.82	4.27	5.65	6.12	6.53	5.77	4.05	3.15	3.20	4.13	5.41	5.02	4.04
	HDD50	1	1	0	0	0	0	0	0	0	0	0	0	0
	HDD65	584	222	141	34	2	0	0	0	0	0	0	22	163
	CDD50	9466	248	306	560	782	1043	1156	1245	1253	1109	907	548	309
	CDD65	4576	4	27	129	334	578	706	780	788	659	442	121	8
	CDH74	66876	144	464	1756	4462	8380	10922	12488	12406	9345	5109	1231	169
	CDH80	41639	25	149	762	2444	5252	7246	8448	8316	5863	2717	389	28
(14) Wind	WSAvg	4.5	4.0	4.4	4.6	5.0	4.9	5.3	4.6	4.3	4.7	4.4	4.1	3.9
(15) Precipitation	PrecAvg	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
	PrecMax	0.4	0.2	0.1	0.2	0.2	0.0	0.0	0.0	0.0	0.0	0.2	0.0	0.0
	PrecMin	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
	PrecStd	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	83.0	92.0	99.2	107.0	110.4	111.7	110.6	111.3	109.2	104.8	93.4	83.5
		MCWB	60.8	64.6	66.7	69.2	71.5	72.3	71.3	73.2	72.7	70.1	66.5	61.4
	2%	DB	77.7	84.2	93.4	102.1	106.6	108.3	108.0	108.4	105.9	99.6	88.1	78.5
		MCWB	58.0	60.5	64.1	67.8	70.2	70.8	70.8	72.3	71.3	69.3	64.4	59.6
	5%	DB	73.6	79.5	88.1	97.6	103.3	105.5	106.1	105.9	102.9	95.9	84.2	74.8
		MCWB	56.4	58.3	62.0	66.0	69.0	69.7	70.3	71.4	70.5	67.8	63.2	57.7
	10%	DB	70.4	75.1	83.5	93.0	99.6	102.6	104.0	103.7	100.0	92.6	81.0	71.9
		MCWB	55.0	56.3	59.6	64.2	67.6	69.0	69.9	70.8	69.5	66.6	61.9	56.5
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	61.7	65.9	68.3	71.1	73.5	74.3	74.5	75.6	75.2	72.8	68.2	63.2
		MCDB	80.5	88.5	94.9	101.9	104.8	106.3	104.2	104.4	104.2	98.7	87.5	80.9
	2%	WB	58.7	61.7	65.1	68.5	71.5	72.3	72.8	74.1	73.1	70.5	65.7	60.2
		MCDB	75.7	82.6	90.4	99.1	102.9	104.1	103.1	103.7	100.4	94.9	84.9	76.4
	5%	WB	56.9	58.8	62.5	66.5	69.9	71.0	71.7	72.9	71.7	68.9	63.8	58.4
		MCDB	72.4	77.2	85.9	95.2	100.3	101.8	102.2	102.3	99.1	92.5	82.1	73.5
	10%	WB	55.2	56.9	60.4	64.6	68.2	69.8	70.7	71.8	70.5	67.5	62.2	56.8
		MCDB	69.7	74.4	81.8	91.6	97.4	100.0	101.0	100.9	97.2	90.3	79.7	71.1
(35) Mean Daily Temperature Range	5% DB	MDBR	26.4	28.1	28.9	30.2	29.5	28.1	27.7	27.1	26.2	25.5	25.9	26.0
		MCDBR	31.2	33.7	36.2	36.6	34.1	31.7	30.3	29.6	29.6	29.4	29.8	29.9
		MCWBR	17.3	16.7	15.8	14.9	13.2	11.0	11.2	10.9	10.7	11.6	14.0	16.2
	5% WB	MCDBR	29.5	32.5	33.7	34.9	33.3	30.1	28.7	27.5	28.3	27.5	28.2	28.2
		MCWBR	17.0	16.9	15.3	14.7	13.5	11.4	11.5	10.9	11.3	11.9	14.2	16.0
(40) Clear-Sky Solar Irradiance	taub		0.388	0.431	0.486	0.587	0.601	0.523	0.507	0.500	0.491	0.478	0.409	0.391
	taud		2.059	1.918	1.783	1.597	1.588	1.765	1.816	1.841	1.851	1.864	2.015	2.049
	Ebn at Noon		269	266	258	235	232	249	252	254	253	247	259	261
	Edh at Noon		46	57	69	86	86	72	68	66	64	60	48	45
(44) All-Sky Solar Radiation	RadAvg		1434	1718	2066	2318	2471	2577	2510	2380	2150	1817	1517	1346
(45)	RadStd		35	65	68	51	47	30	25	26	47	29	27	32

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (2 neighbors)	+0.54	N/A	N/A	N/A	-1.71	-3.06	N/A	N/A	N/A	N/A

Nomenclature: See separate page